05- Bit Manipulation

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Homework - Correct answer:

A → 10 B → 11 C → 12 D → 13 E → 14 F → 15

0x 4C 75 4E 61 54 69 43

0100 1100 → 12 → L
0111 0101 → 21 → U
0100 1110 → 14 → N
A T I C

	4 bits MIN/MAX		8 bits MIN/MAX		16 bits MIN/MAX	
SIGNED INTEGER	−8	7	−128	127	−32768	32767
	1000	0111 $-2^3 2^2 2^1 2^0$	1000 0000	0111 1111	1000 0000 0000 0000	0111 1111 1111 1111
UNSIGNED INTEGER	0	15	0	255	0	65 535
	0000	1111 $2^3 2^2 2^1 2^0$	0000 0000	1111 1111	0000 0000 0000 0000	1111 1111 1111 1111

$-2^7\ 2^6\ 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0$

255 ←

1 1 1 1 1 1 1 1

+ 127 + 64 + 32 + 16 + 8 + 4 + 2 + 1

1

256 − 1 = 255

1 0 0 0 0 0 0 0 0

8192 4096 2048 1024 512 256 128 64 32 16 8 4 2 1

+ 1 1 1 1 1 1 1 1 1 1 1 1 1 = 8191

1

1 0 0 0 0 0 0 0 0 0 0 0 0

Turn off every bit

$x = \emptyset \rightarrow$ 0000 0000 0000

Turn on every bit

$x = -1 \rightarrow$ 1111 1111 1111

Turn on one specific bit

	0 0 0 0 0 0 1	$x = 1$
1 << 1	0 0 0 0 0 1 0	$x = 1 \times 2$
1 << 2	0 0 0 0 1 0 0	$x = 1 \times 2 \times 2 = 4$
1 << 3	0 0 0 1 0 0 0	$x = 1 \times 2 \times 2 \times 2 = 8$
	⋮	
1 << 6	1 0 0 0 0 0 0	$x = 1 \times 2^6 = 64$

<< left-shift operator

Result

6 << 2 ?

16 8 4 2 1

1 1 0 0 0 = 24

>> right-shift

$100_{10} \longrightarrow 1100100_2$

100 >> 1 = 1100100 = 50

32 16 8 4 2 1

100 >> 2 = 1100100 = 25

16 8 4 2 1

100 >> 3 = 1100100 = 12

8 4 2 1

~ (bitwise negation)

~ (110 1101)

001 0010

~0 → 1

~1 → 0

& bitwise and

1 & 1 = 1

1 & 0 = 0

0 & 1 = 0

0 & 0 = 0

23 & 35

35

23 → 010111 &

35 → 100011

000011 → 3

| bitwise or

(|) bitwise or

$$\begin{array}{r} 1\,1 \\ \hline 1 \end{array} \quad \begin{array}{r} 1\,1 \\ 0 \\ \hline 1 \end{array} \quad \begin{array}{r} 1\,0 \\ 1 \\ \hline 1 \end{array} \quad \begin{array}{r} 1\,0 \\ 0 \\ \hline 0 \end{array}$$

(^) xor

$$\begin{array}{r} 1\,\wedge\,1 \\ \hline 0 \end{array} \quad \begin{array}{r} 1\,\wedge\,0 \\ \hline 1 \end{array} \quad \begin{array}{r} 0\,\wedge\,1 \\ \hline 1 \end{array} \quad \begin{array}{r} 0\,\wedge\,0 \\ \hline 0 \end{array}$$

Solve the following :
Write the result as base 10

a) 17 & 12
b) 7 << 2
c) (~(1<<1 | 1<<3 | 1<<5)) & 29
d) 17 | 12
e) (~(13 >> 2)) & 39

(~ (13 >> 2)) & 39

$13_{10} = 1101_2$
$39_{10} = 100111_2$

1101 >> 2 = 11 ~~01~~

~(0000000011)
& ...111111110 0
& ...000100111
...000100100 = 36_{10}
32 16 8 4 2 1